



- gdn** GatedDeltaNet, $O(L)$ linear attention, **strictly causal** — 12/18
- swa** sliding window, band ± 128 (257 wide), bidirectional — 4/18
- full** global softmax, bidirectional, *the only blocks that receive the pair bias* — 2/18, at indices 7 and 15

Measured [F103]. Scrambling the right half of the input leaves the left half of a **gdn** block's output changed by **exactly** 0.000e+00; the same test moves **swa** by 2.0×10^{-1} and **full** by 1.4×10^{-1} . Two thirds of the trunk is therefore left-context-only. A nucleotide can be paired with a partner more than 128 nt to its *right* only through the two **full** blocks — and those are the blocks finding 92 measured at chance (AUROC 0.44–0.52) against the contact map. PHAROS is an encoder, not a language model, so the causal mask buys nothing it needs.

Figure 1: t